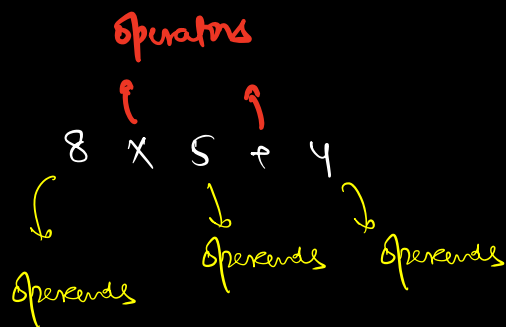


⇒ Notations

+	Infix	→	$8 \times 5 + 4$
+	Postfix	→	$8 \ 5 \times$
+	Prefix	→	$\times \ 8 \ 5$



⇒ Infix ⇒ Operand **operator** operand → $a + b$

⇒ Postfix ⇒ Operand Operand **operator** → $a \ b \ +$

⇒ Prefix ⇒ **operator** Operand Operand → $+ \ a \ b$

$$\Rightarrow \quad 8 \times 5 + 4 \Rightarrow 44$$

~~72~~

→ Operator precedence ⇒ $\frac{\times, /}{+, -}$

⇒ $10 + 3 \times 4 - 2 \rightarrow$ infix → order of operators are not in the same as order of their execution

⇒ $10 + 12 - 2$

→ left to right →

$$\Rightarrow 22 - 2$$

$$\Rightarrow \underline{\underline{20}}$$

prefix

$$8 \times 5$$

$$\Rightarrow X85$$

postfix

$$8 \times 5$$

$$\Rightarrow 85X$$

Imp to postfix

$$A + \underbrace{B \times C}$$

$$\Rightarrow A + BCX$$

$$\Rightarrow \underline{\underline{A BCX +}}$$

$$\Rightarrow 10 + \underbrace{3 \times 4} - 7$$

$$\Rightarrow 10 + \underbrace{34X} - 7$$

$$\Rightarrow \underbrace{10 34X +} - 7$$

$$\Rightarrow 10 34X + 7 -$$

$$\Rightarrow \underbrace{(10 + 3)} \times 2 - \underbrace{(7 - 6)} \times \underbrace{(4 + 8)}$$

$$103+ \times 2 - 76- \times 48+$$

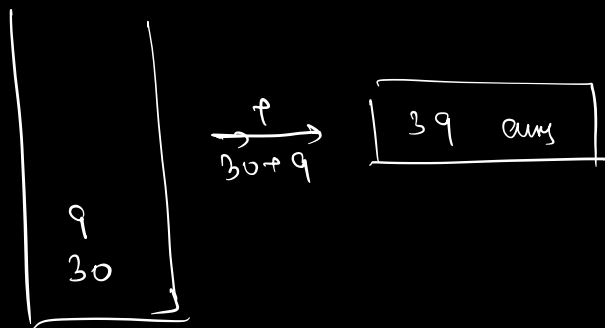
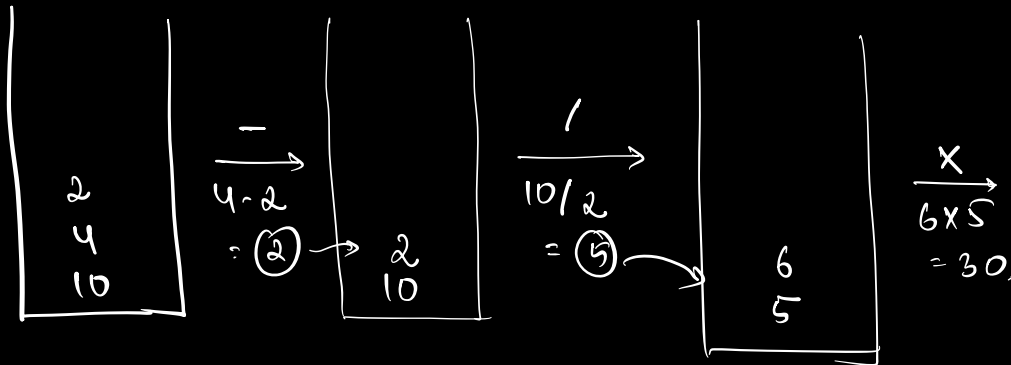
$$\begin{array}{c} \searrow \quad \swarrow \quad \searrow \quad \swarrow \\ 103+2X - 76-48+X \end{array}$$

$$\downarrow \quad \downarrow$$

$$[103+2X 76-48+X -]$$

Q. Given a postfix notation find its value

$$\Rightarrow 1042 - \underset{\substack{\uparrow \\ abx}}{6 \times 9 \div} = 1042 - \underset{abx}{6 \times 9 \div}$$



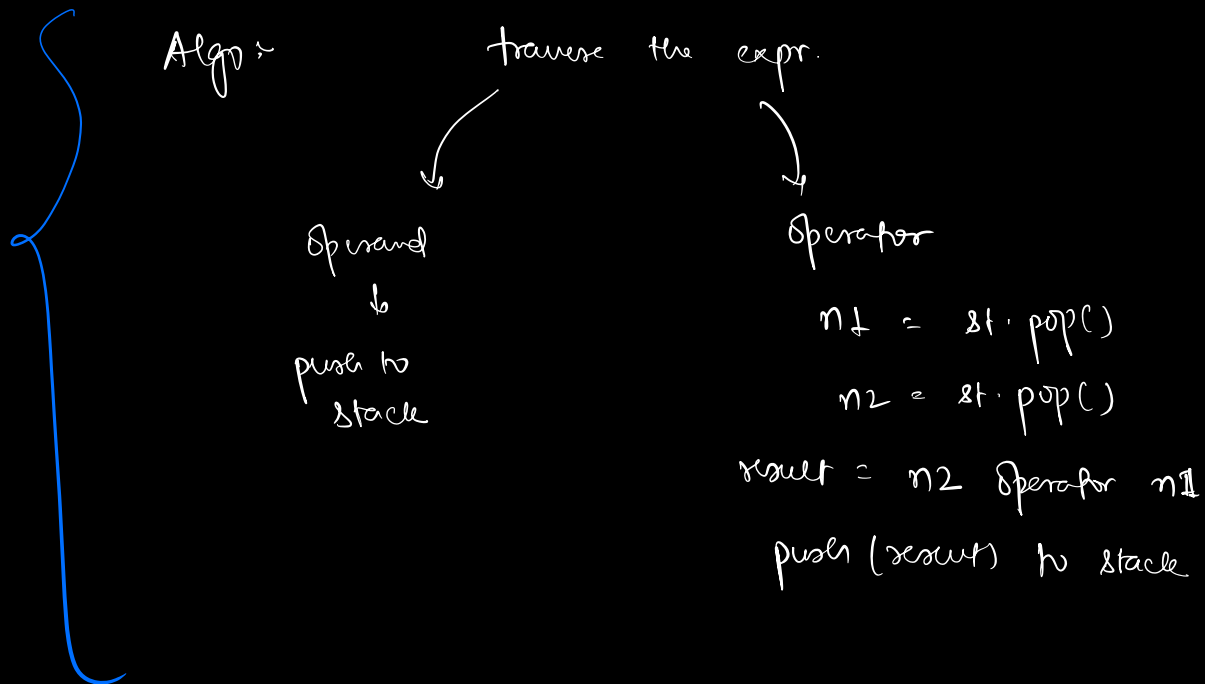
$$\Rightarrow 10 \mid \underbrace{(4-2) \times 6 + 9}$$

$$10 \mid 42 - x6 + 9$$

$$1042 - (\times 6 + 9)$$

$$1042 - 16x + 9 \Rightarrow 1042 - 16x + 9 +$$

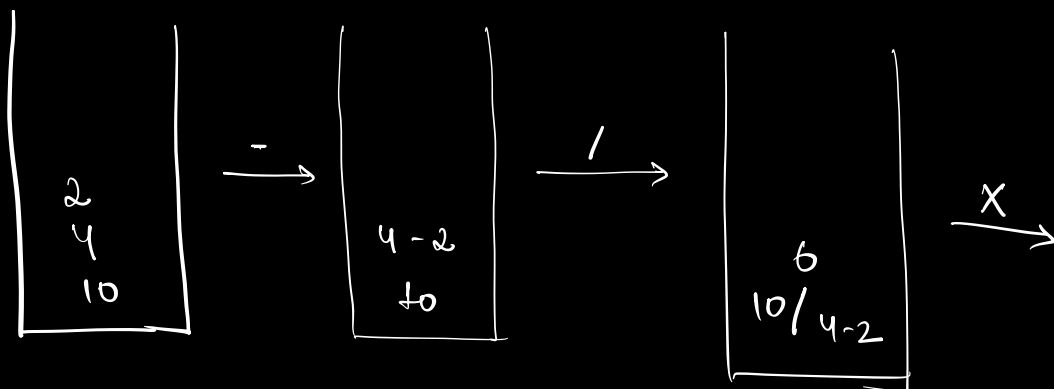
Postfix $\rightarrow$ Operators come in the order of their execution



: Postfix to Infix

Q Given a postfix notation find its value

$$\Rightarrow 10 \ 4 \ 2 \ - \ 6 \times 9 \ +$$



$$\boxed{} \xrightarrow{+} [10 / 4 - 2 \times 6 + 9]$$

postfix infix $\longrightarrow$ ✓
 prefix ↓

postfix $\Rightarrow$ $\text{---} + \text{---} \times \text{---}$

infix $\Rightarrow$ $\text{---} + \text{---} \times \text{---}$
 $\downarrow$
 $()$

ex $\Rightarrow$ postfix $\Rightarrow$ $8 \ 5 \ \underline{+} \ 10 \ \times$
 $\Rightarrow$ $13 \ 10 \ \times$
 $\Rightarrow$ 130

How

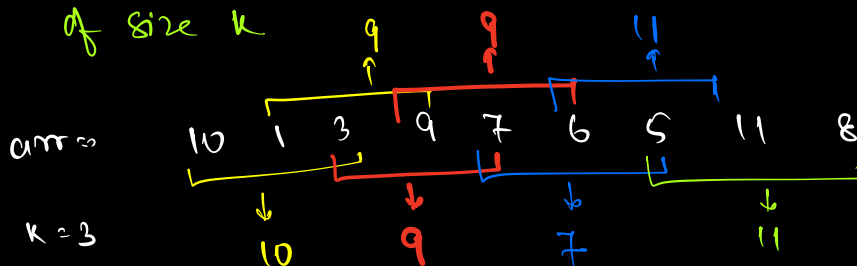
infix $\Rightarrow$ $(8 + 5) \times 10$

$$= 13 \times 10$$

$$= 130$$

! Sliding Window max :-

Given an arr[N], k, find the maxes for all windows of size k



BF $\Rightarrow$ for every subarray of size k . find max

$$TC \Rightarrow O[(N-k+1) \times k]$$

$$k=1$$

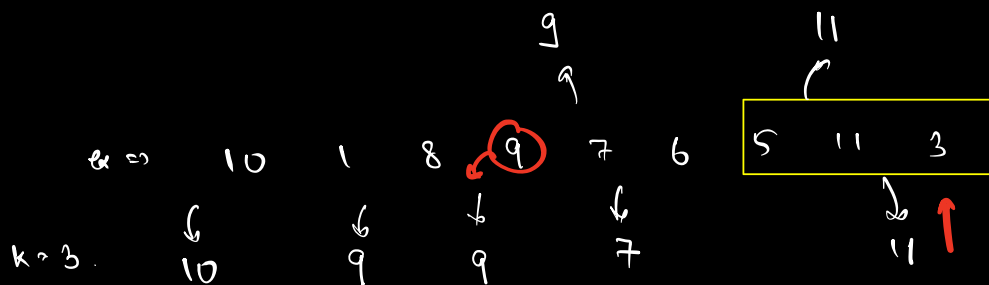
$$O(N)$$

$$k=N/2$$

$$O(N^2)$$

$$k=N$$

$$O(N)$$

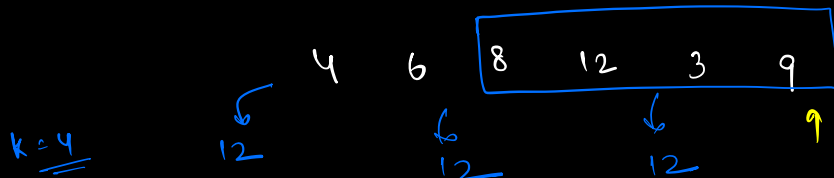


~~10~~ ~~1~~ ~~8~~ ~~9~~ ~~7~~ ~~6~~ ~~5~~ 11 3

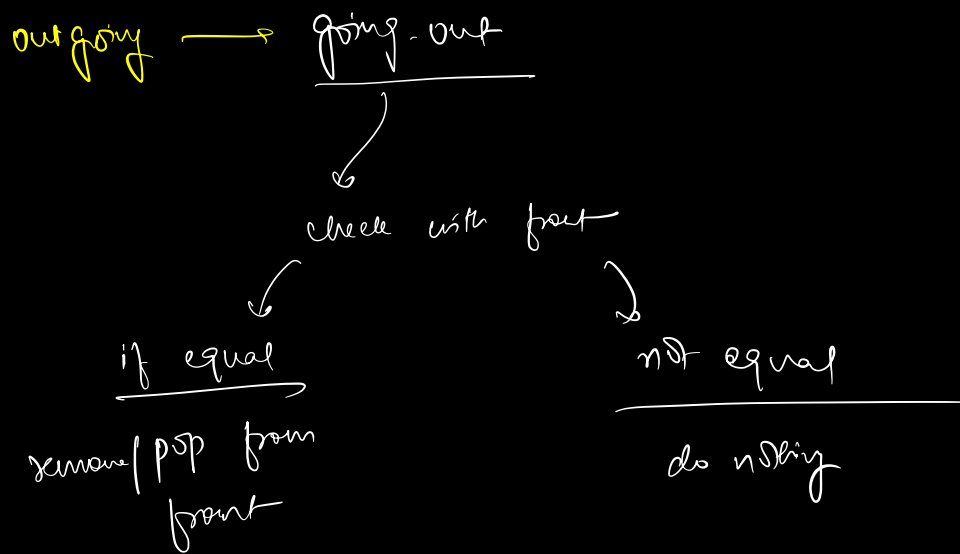
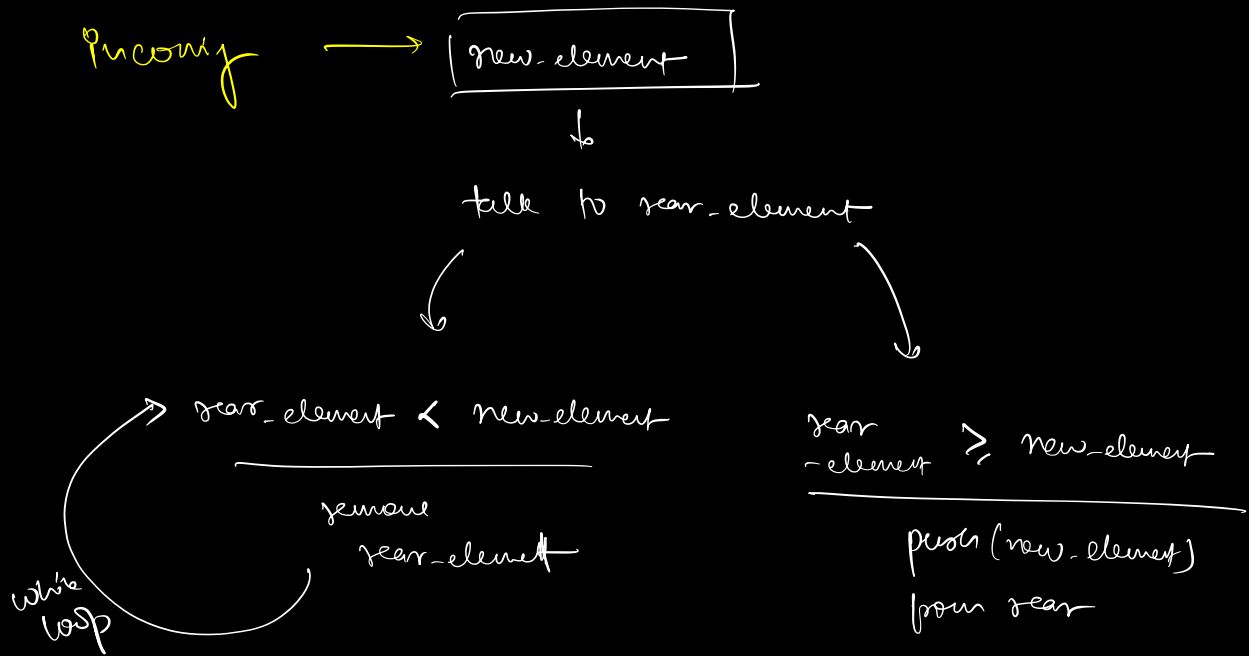
front

rear

$$\Rightarrow an \sim [10, 9, 9, 9, 7, 11, 11]$$



~~4~~ ~~6~~ 8 12 3 9



max $\rightarrow$ front of queue

$\rightarrow$ operation

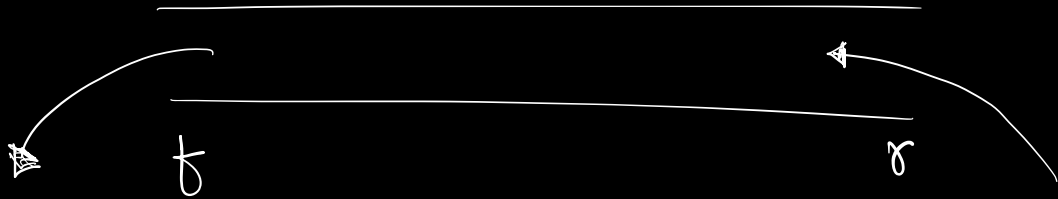
- i) accessing the rear
- ii) " " front

- iii) inserting/removing from rear

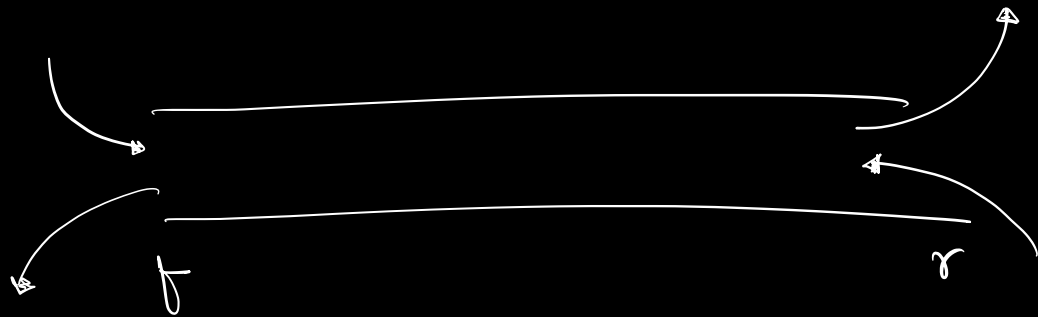
- iv) removing from queue

⇒ Deque → Double Ended Queue

normal
queue



double
ended
queue
(Deque)



→ Impl. array

(-1)
f r

	front	rear
remove →	f++	r--
insert →	b--	r++
access →	arr[f+1]	arr[r]

→ How → code for deque

→ Empt LL → Dll (doubly ll)

DO

→ insert at rear → $O(1)$

→ removal from rear → $O(1)$

→ insert at front → $O(1)$

→ removal at front → $O(1)$

How - implement the code

→ PS → 3

[Strings
HM
op(LL)
✓✓]

→ 29th Oct (Thurs)
9 PM.

$$\Rightarrow -(-9 + 8 \times 9 - 8)$$

$$\begin{array}{c} \downarrow \\ -9 + 8 \times 9 - 8 \quad - \end{array}$$

$$\begin{array}{c} \downarrow \\ -9 + 8 \times 9 - 8 \quad - \end{array}$$

$$-9 + 8 \times 9 - 8 \quad -$$

$$[-9 + 8 \times 9 - 8]$$